FINAL PROJECT · ARTIFICIAL INTELLIGENCE

What the market *will ask next*

Forecasting food commodity prices in Cambodia — from World Food Programme price monitoring to a web app and a Telegram bot.

46 COMMODITIES · 76 MARKETS · XGBOOST, PORTED TO TYPESCRIPT

An early warning *only works if it arrives*

PROBLEM

A price rise lands hardest on the people with the least slack.

- For a low-income household food is the largest monthly cost, so a price rise is not an inconvenience.
- WFP tracks these markets to catch that early, but publishes what prices were — not what they will be.
- And the record itself ships as a 85,107-row CSV: readable by analysts, nobody else.

→

SOLUTION

Make the answer reachable in two clicks or one message.

- 3,442 price series across 46 commodities and 76 markets, charted for anyone with a browser.
- A next-period forecast on each, with the line between recorded and predicted drawn honestly.
- Free to run — so a ministry, an NGO or a co-op could stand the same thing up on its own feed.

A demonstration on a snapshot, not a deployed service — the data ends March 2026 and the app says so, series by series.

THE DATA

Twenty-three years *of market visits*

SOURCE	WFP Global Food Prices — Cambodia (HDX)
OBSERVATIONS	85,107
RANGE	2003-01-15 to 2026-03-15
COMMODITIES	50 raw
MARKETS	86 raw
TARGET	usdprice — price per unit in USD

FILTERING

`85,107 rows`
everything WFP has recorded

↓

`from 2019 onward`
older reporting is sparse and pre-dollarisation

↓

`commodities with 300+ rows`
a series needs history to have a lag

↓

`67,654 modelled rows`
46 commodities · 76 markets · 4,035 series

FRAMING IT AS A LEARNING PROBLEM

The split is the *whole experiment*

Each (commodity, market, pricetype) forms a time series. The model predicts the next observation in that series from its own recent history.

THE TRAP

A random train/test split lets the model learn from June to predict April. Prices are autocorrelated, so it scores brilliantly and forecasts nothing. The split has to be chronological.

CHRONOLOGICAL HOLD-OUT

Cut point	2025-04-15 (85th percentile of dates)
Train	57,878 rows · everything before the cut
Test	9,776 rows · everything after
Never seen	the test period, by any model, at any point

FEATURE ENGINEERING

Ten features, built *inside each series*

PRICE HISTORY

`lag_1`
the previous reported price

`lag_3`
the price three reports back

`rolling_mean_3`
mean of the last three

CALENDAR

`month`
1-12, seasonality of harvest

`quarter`
1-4, coarser seasonality

IDENTITY

`commodity_enc`
which food

`market_enc`
which market

`admin1_enc`
which province

`category_enc`
cereals, meat, pulses ...

`pricetype_enc`
retail or wholesale

> Lags are computed within each series (grouped shift), never across them — a rice price in Battambang can never leak into a pork price in Phnom Penh. Rows without three prior observations are dropped rather than imputed.

MODEL COMPARISON

Beating "tomorrow *equals today*"

The naive baseline — predict the last price — already scores R^2 0.92, because prices are sticky. It is the number any model has to beat to be worth deploying.

MODEL	MAE (USD)	RMSE (USD)	R^2
Naive baseline (lag_1)	0.1799	0.4088	0.9239
Random Forest (reproduced)	0.1690	0.3352	0.9489
XGBoost (final model)	**0.1642**	**0.3295**	**0.9506**

Random Forest reproduced locally from the notebook's settings (n=300, depth 15, seed 42).
XGBoost figures come from the trained price_model.pkl itself, evaluated on the held-out period.

RESULT

The model earns *its place*

$0.164

MEAN ABSOLUTE ERROR

on held-out 2025-26 prices

0.951

R^2 ON THE TEST PERIOD

variance explained

8.7%

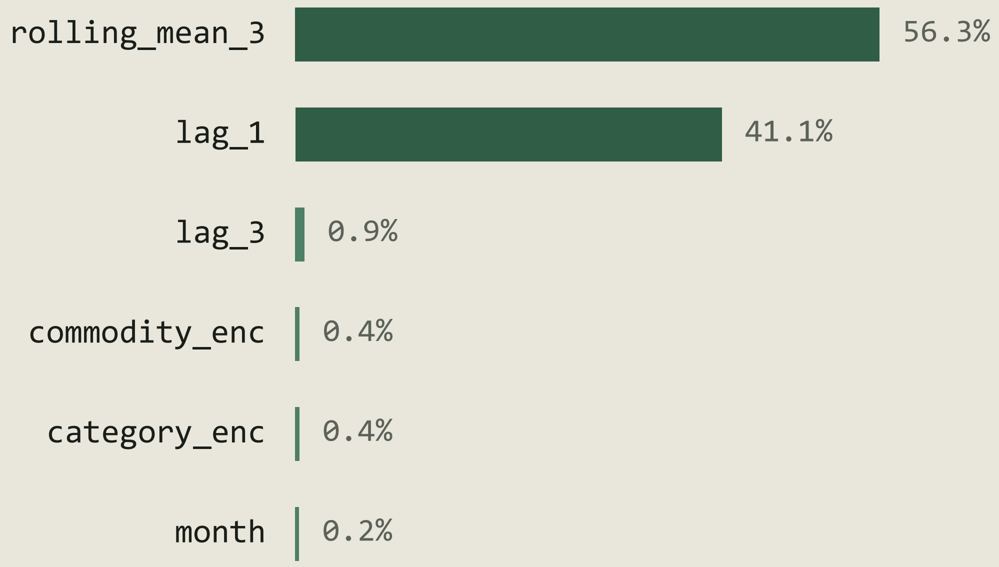
BETTER THAN NAIVE

MAE reduction vs. last price

An average miss of about sixteen US cents on a basket whose prices mostly sit between $0.30 and $3.00 — accurate enough to plan around, not precise enough to trade on. Both halves of that sentence matter.

INTERPRETATION

It learned to be *a very good smoother*



THE HONEST READING

97% of the model

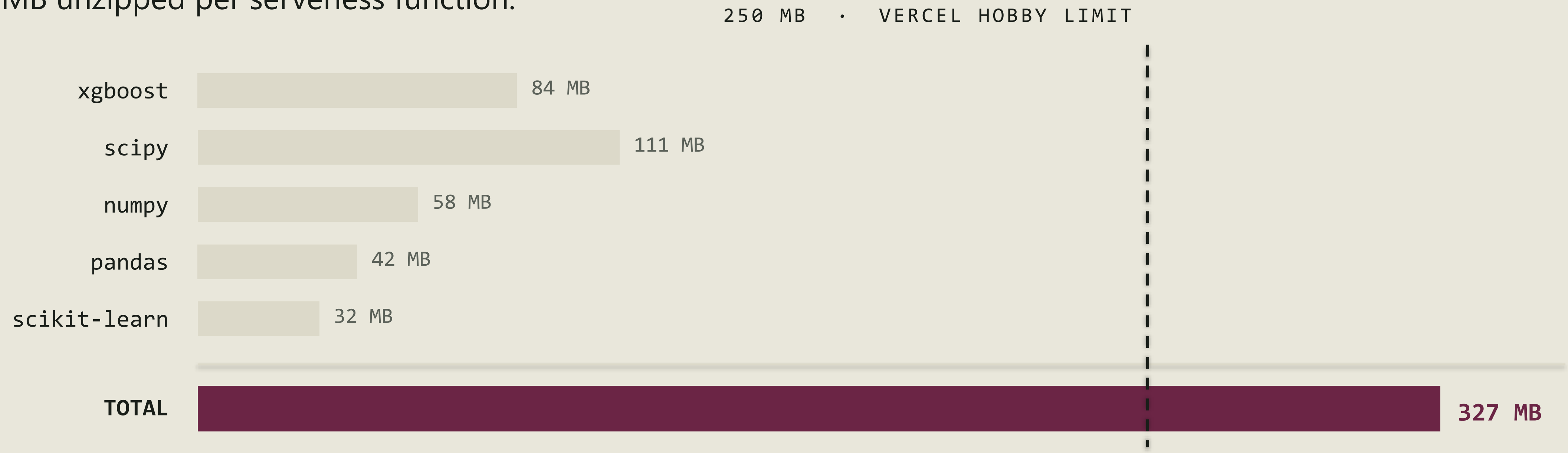
rests on two features: the recent mean and the last price.

Commodity, market, province and season together account for barely two percent. This is a well-calibrated smoother, not a market oracle — and saying so is part of the result.

Gain-based feature importance from the trained booster.

The model fit.
The libraries didn’t.

A notebook that only runs in Colab isn't a product. Deploying to Vercel's free tier meant meeting a hard limit: 250 MB unzipped per serverless function.



The deploy fails before a single prediction is ever served.

THE INSIGHT

A boosted tree is *just a lot of if-statements*

Training needs XGBoost. Predicting does not. Inference is: walk each tree comparing one feature to one threshold, sum the leaf weights, add the base score. Fifteen lines in any language.

400	10	2.4 MB	0
trees traversed	features each	of JSON	ML dependencies

The model exports to JSON: tree arrays, encoder maps, feature order, and a 24-point price history per series. The deployed function ships none of the Python stack.

```
function rawPredict(features: number[]): number {
  let total = BASE_SCORE
  for (const tree of TREES) {
    let node = 0
    while (tree.l[node] !== -1) {
      node = features[tree.f[node]] < tree.t[node]
        ? tree.l[node]
        : tree.r[node]
    }
    total += tree.w[node]
  }
  return total
}
```

VERIFICATION

Two languages, *one number*

A port is only worth anything if it agrees with the original. The test suite pins the TypeScript traversal to the Python model's own answer for a known series — a wrong feature order or a flipped comparison changes that number, and nothing else would catch it.

19 tests cover the traversal, the fuzzy matching, the commodity/market splitter and every failure path.

REFERENCE CASE

commodity	Rice (mixed, low quality)
market	Phnom Penh · Wholesale
last recorded	2022-06 · $0.44
python xgboost	0.460955
typescript port	**0.4609556**

agree to 1e-5 · asserted on every test run

THE RESULT

It runs
in a browser

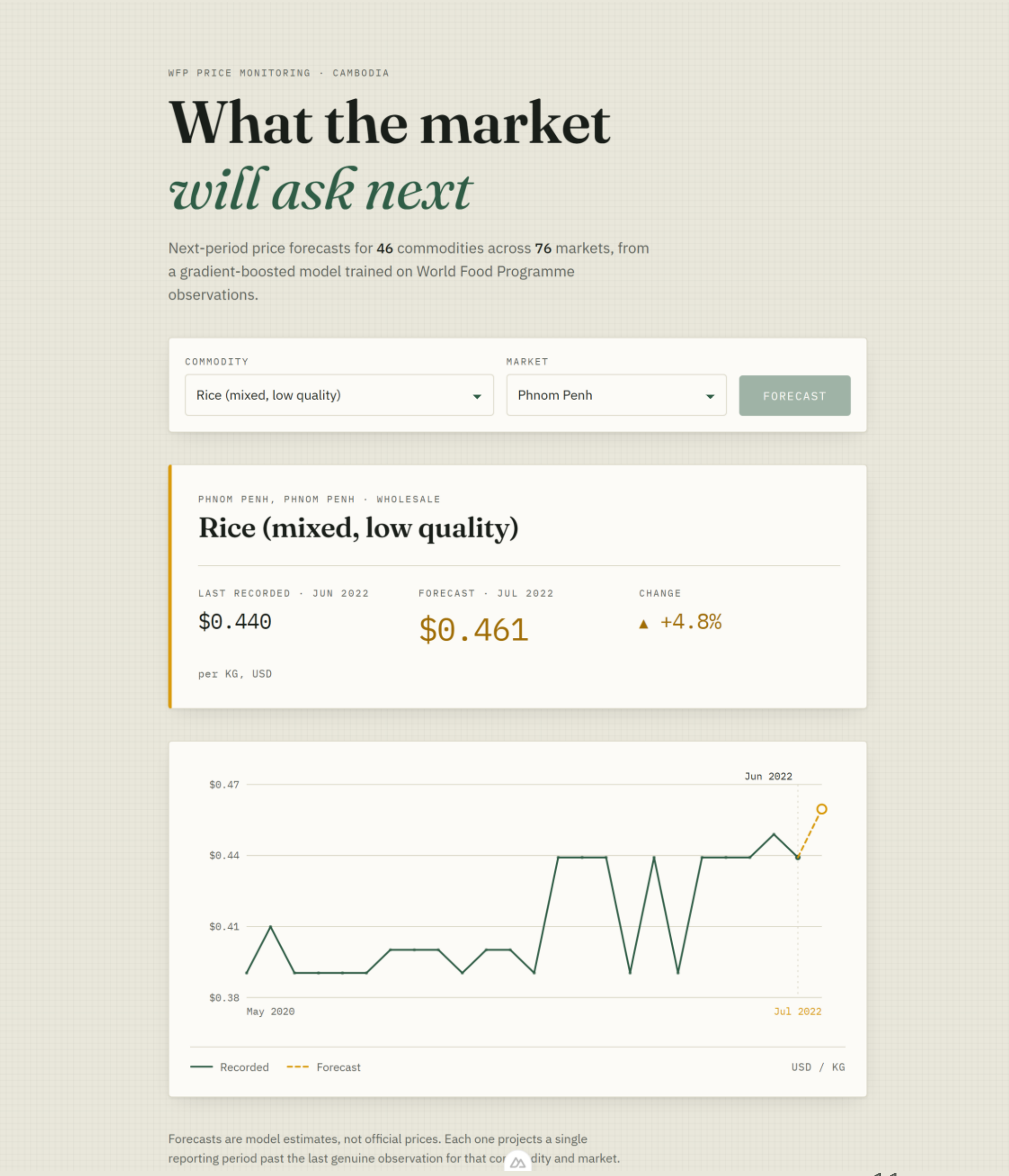
Nuxt 4 on Vercel. The default forecast is server-rendered, so the chart is on screen at first paint — no spinner, no empty state.

Dependent dropdowns
picking a commodity narrows the markets to those that carry it

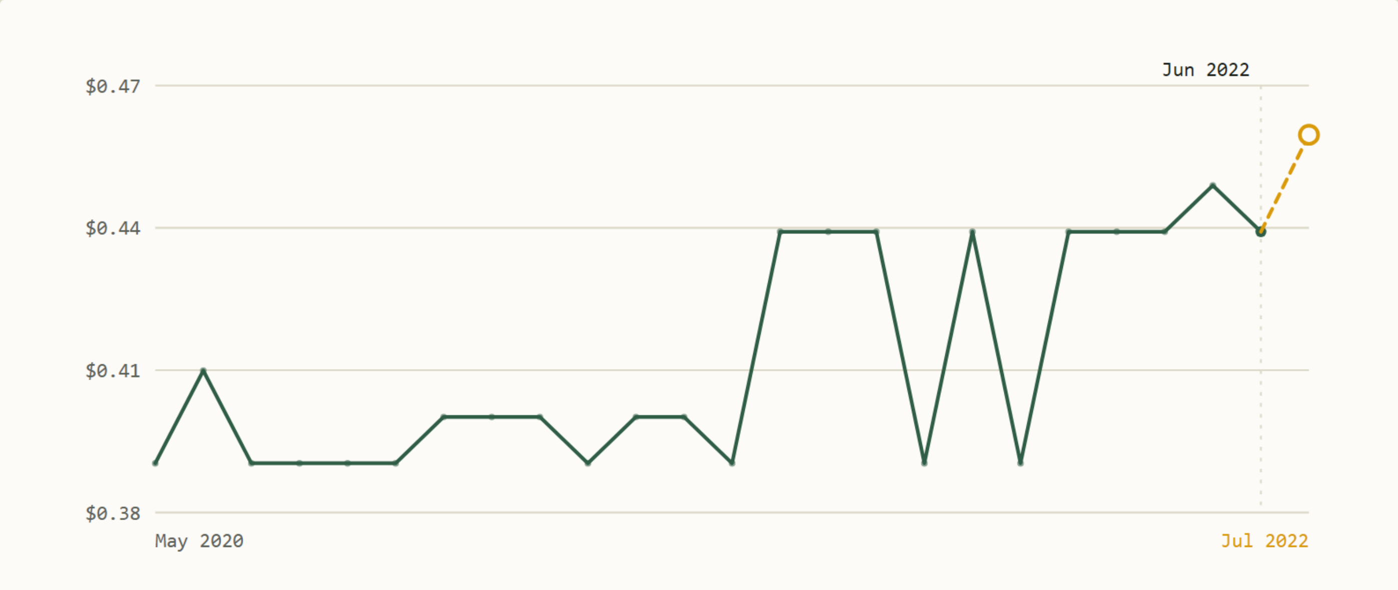
Fuzzy input
"rice" resolves to Rice (mixed, low quality) — shortest match wins

Cached at the edge
options for a day, forecasts for an hour

Responsive to 390px
the chart switches to a narrower viewBox, not smaller type



Where the record ends *must be unmistakable*



A charting library would draw the forecast as the next point on the same line. That is a lie of omission: it renders a model estimate with exactly the same authority as a recorded observation.

Solid green
prices actually recorded

Dashed amber
the model's estimate, rising

Dashed red
the model's estimate, falling

Hollow ring
not an observation

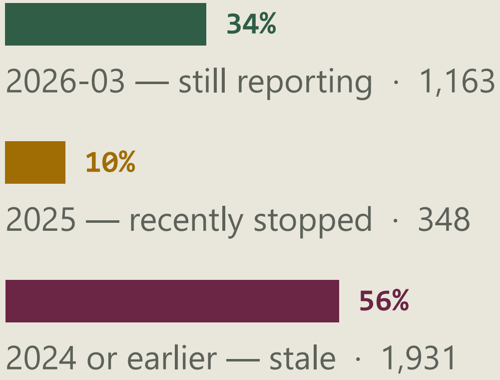
Hand-rolled SVG — about 200 lines, no charting library.

FORECAST HORIZON

One period past the record, *not one month past today*

The model forecasts one reporting period past each series' own last observation — not one month past today. Series stopped reporting at very different times, so that date differs for every commodity and market.

LAST OBSERVATION ACROSS 3,442 SERVED SERIES



SO IS IT ACTUALLY FORECASTING?

Yes — and that is what the test period measures.

Training stopped at 2025-04-15; all 9,776 test rows fall after it. The model was scored on dates it had never seen — a real out-of-sample forecast.

What it cannot do is forecast today from a series that went quiet in 2022 — and the app prints the real date rather than hiding it.

Rice at Phnom Penh, wholesale: last recorded June 2022, so the app labels its forecast July 2022 — not next month.

SECOND SURFACE

The same engine,
in a chat window

One prediction engine, two front doors. The bot shares the exact code the web app calls — a Telegram update in, a formatted forecast out.

Reply in the response body

the webhook returns the sendMessage call itself — no outbound request, and the bot token never reaches Vercel

Always HTTP 200

a non-2xx makes Telegram retry the same update forever

Sparkline in Unicode blocks

the only chart a chat window can render inline

```
/predict rice phnom penh
```

Rice (mixed, low quality)

Phnom Penh, Phnom Penh · Wholesale

Last recorded (2022-06): $0.440 / KG

Next period forecast: $0.461

▲ +4.8%

2020-05 → 2022-06, 24 observations

What this model *cannot do*

One period ahead, and no further

Every feature is a recent price. Feed the model its own output and error compounds fast — there is no multi-step forecast here.

It cannot see shocks coming

Fuel costs, export bans, floods, festival demand. None are in the feature set, and they are exactly what moves prices sharply.

Stale series forecast from stale prices

A series last reported in 2022 gets a 2022-shaped answer. Honest, but not useful for today.

No uncertainty interval

A single number, not a range. The dashed line signals doubt visually, but does not quantify it.

IF I KEPT GOING

Three things *I would add next*

Prediction intervals

Quantile regression at 10/90 to turn the dashed line into a shaded band — the honest version of the same signal.

Exogenous features

Fuel price, rainfall, and the lunar calendar for festival demand. The categorical features contribute two percent; these might not.

Retraining on a schedule

The export script already runs headless. A monthly job against the live HDX feed would keep every series current.

THANK YOU

Questions *welcome*

DATA	WFP Global Food Prices — Cambodia
MODEL	XGBoost · 400 trees · scikit-learn pipeline
RUNTIME	Nuxt 4 · Nitro · TypeScript tree traversal
SURFACES	Web app · Telegram bot · one shared engine